8 The variation with time t of the displacement y of a wave A, as it passes a point P, is shown in Fig. 8.1.

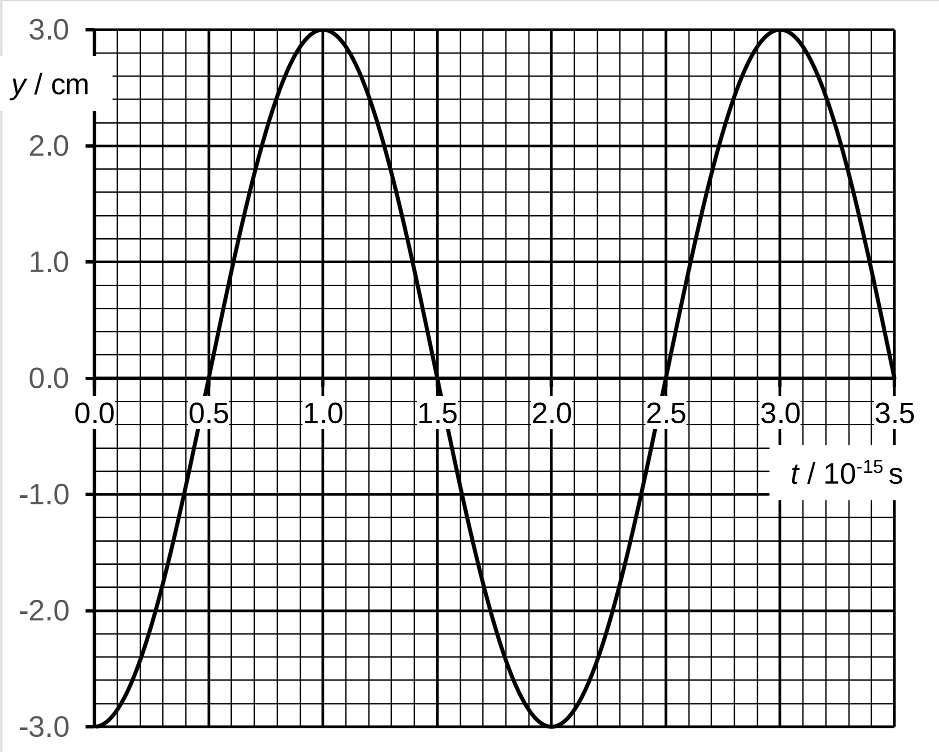

Fig. 8.1

The intensity of wave A is $2I$.

(a) Use Fig. 8.1 to determine the frequency of wave A.

From graph, $T = 2.00 \times 10^{-15}$

$f = 1/T = 1 / 2.00 \times 10^{-15} = 5.00 \times 10^{14}$ Hz

frequency = ... Hz [2]

(b) A second wave B with the same frequency and speed as wave A also passes point P. Wave B has intensity I and is lagging wave A by $\frac{\pi}{2}$ rad.

On Fig. 8.1, sketch the variation with time t of the displacement y of wave B. Label the graph B.

Show your working.

If intensity of $2I$ has amplitude of A, then intensity of I will have amplitude of $(1/\sqrt{2})$ A.

Amplitude of wave B = $(1/\sqrt{2})(3.0) = 2.1$ cm.

sketch a graph with shape of $-\sin \omega t$ with amplitude of 2.1 cm.

(c) Explain why wave A and wave B are coherent.

They have a constant phase difference of π/2 rad between them.

(d) Wave R is the resultant wave due to the superposition of wave A and wave B. An amplitude of wave R occurs at $t = 1.2 \times 10^{-15}$ s.

Using your answer in **(b)** and Fig. 8.1, determine, in terms of I, the intensity of wave R at point P.

From graph, at $t = 1.2 \times 10^{-15}$ s,

Amplitude of wave R = displacement of wave A + displacement of wave B = 2.4 + 1.3 = 3.7 cm

Intensity of wave R = $(3.7/3.0)^2$ times intensity of wave A = $(3.7/3.0)^2\, 2I = 3.0\, I$

intensity = ... [2]

(e) After passing through point P, wave R is plane-polarised by passing it through a polarising filter. Determine, in terms of I, the intensity of the plane-polarised wave R.

Intensity of plane-polarised wave R = ½ (Intensity of unpolarised wave R) = ½ (3.0 I) = 1.5 I

intensity = ... [1]

(f) Wave R is made to pass through a double slit in an interference experiment, as shown in Fig. 8.2.

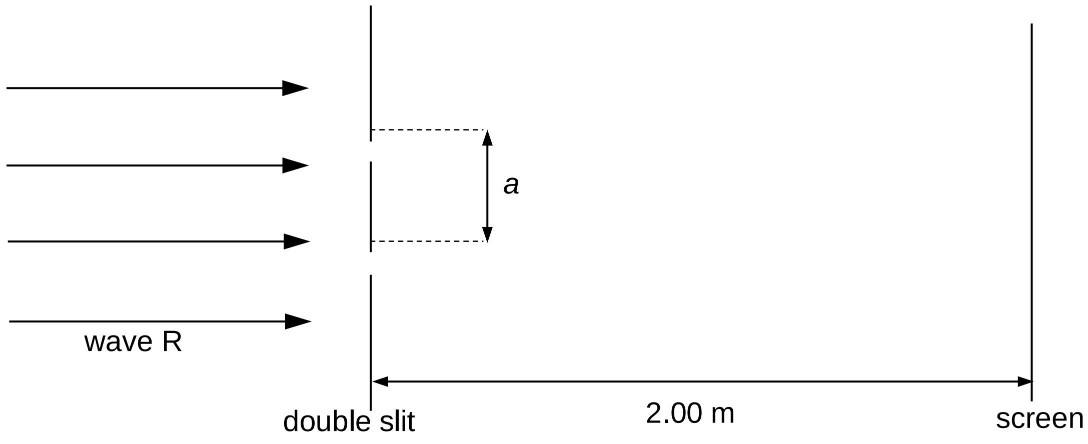

Fig. 8.2

The separation between the slits is *a*. The fringes are viewed on a screen at a distance 2.00 m from the double slit. The fringe separation *x* is measured for different slit separation *a*. A graph of *a* against $\frac{1}{x}$ is shown in Fig. 8.3.

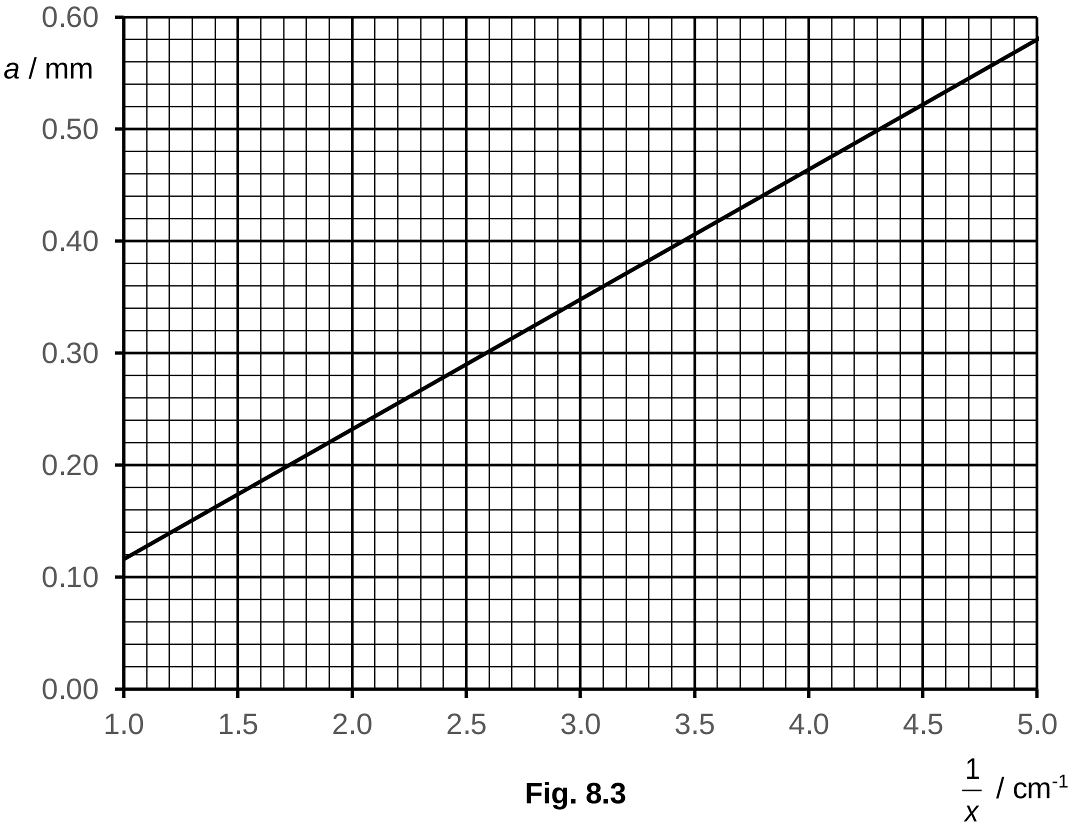

Fig. 8.3

(i) Determine the wavelength of wave R.

From graph, Gradient $= \dfrac{(0.50-0.14)\times10^{-3}}{(4.30-1.20)\times10^{2}} = 1.16\times10^{-6}$

$x = \lambda D / a \;\rightarrow\; a = \lambda D\,(1/x)$

Therefore, λD = Gradient

λ = Gradient / D = 1.16×10^{-6} / 2.00 = 5.8×10^{-7} m (accept 580 – 600 nm)

(ii) State the effect, if any, on the appearance of the fringes observed on the screen when the following changes are made separately, at a fixed value of *a*:

1. wave R is replaced by a blue laser light,

Wavelength of blue light is shorter, so fringe separation will appear closer together.

2. the width of each slit is increased but the separation remains constant.

More light goes through, so intensity of bright fringe will increase, increasing contrast between bright and dark fringes.

Also, diffraction will be less significant / less diffraction effect, (resulting in less overlapping for interference), hence fewer fringes will be observed on the screen.

(iii) At a particular value of slit separation *a*, the variation with distance along the screen of the intensity of the image on the screen is shown in Fig. 8.4.

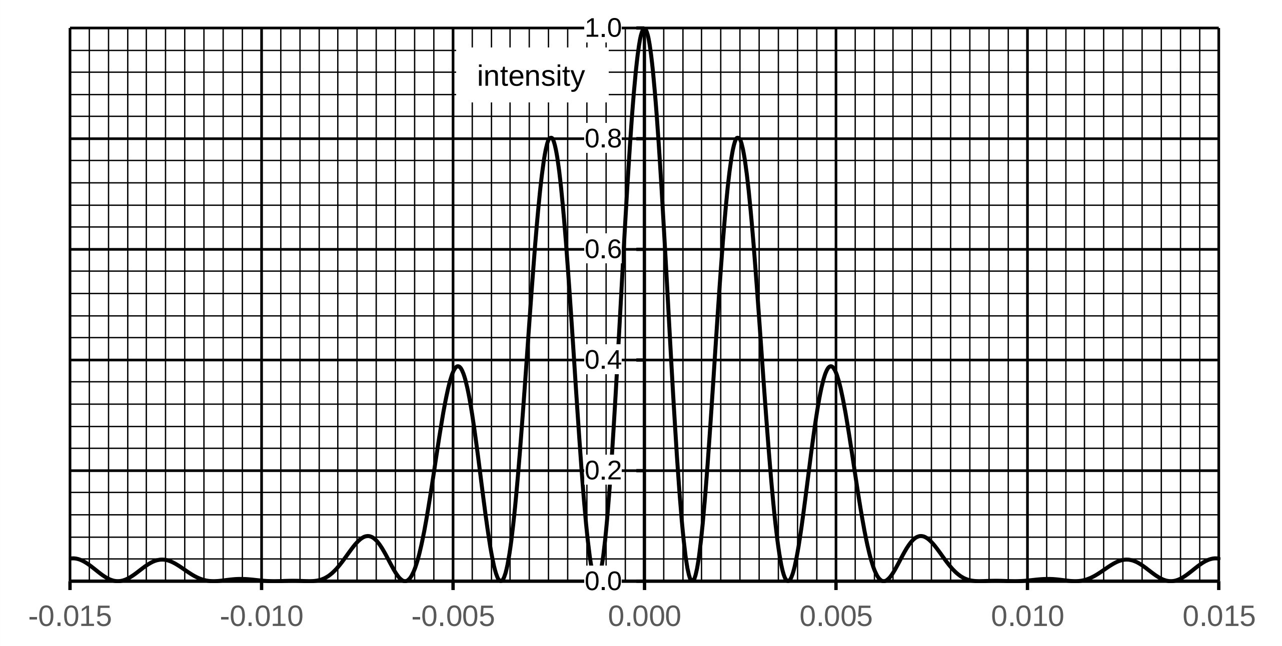

Fig. 8.4

1. Using Fig. 8.4 and Fig. 8.3, determine the value of *a*.

From Fig. 8.4, x = [0.0073 – (– 0.0073) / 6 = 2.43×10^{-3} m

From Fig. 8.3, for 1/x = 1 / 2.43×10^{-3} = 412 m^{-1} = 4.12 cm^{-1}

Reading off Fig. 8.3, a = 0.48 mm (accept 0.47 – 0.49 mm)

2. Using Fig. 8.4, determine the width of each slit.

From Fig. 8.4,

w, width of central maximum = [0.0095 – (– 0.0095) = 0.019 m (accept 0.0085 to 0.0100)

$\tan \theta = \frac{1}{2} w / D$

Also, due to single slit diffraction, $\sin \theta = \lambda / b$

By small angle approximation, $\sin \theta \approx \tan \theta$

$\frac{1}{2} w / D = \lambda / b$

$b = \lambda D / \frac{1}{2} w = (5.80 \times 10^{-7})(2.00) / \frac{1}{2}(0.019) = 1.2 \times 10^{-4}$ m (accept $1.2 - 1.4 \times 10^{-4}$ m)

9 Two metal plates X and Y are contained in an evacuated container and are connected